Aftercare Loop

Helping patients understand their care after the visit, and follow through on it.

Understand → Ask → Verify → Act → Follow Up

Team HW1 · Hackathon pitch

Demonstration using fictional patient data. Not medical advice.

THE PROBLEM

40–80%

of medical information is forgotten right after a visit. Almost half of what is remembered is wrong.

49%

of recommendations recalled accurately one week after a specialty visit

38% vs 65%

recall for patients without a high-school diploma vs college graduates. Teach-back by providers was “rare.”

Sources: Kessels, J R Soc Med 2003 (review; underlying data Anderson et al. 1979) · Laws et al., PLoS ONE 2018

WHO WE SERVE

Underserved women carry the cost of a forgotten plan

Rural, caregiving, cost-constrained, often managing more than one condition. The after-visit summary was never designed for her.

5.8M

US residents delayed medical care in one year because they lacked transportation

9.1%

of women did not take medication as prescribed to save money, vs 7.0% of men

61%

of 53 million US family caregivers are women, averaging 23.7 hours of care a week

12%

of US adults have proficient health literacy; 30% of Medicaid recipients are at the lowest level

Sources: Wolfe et al., Am J Public Health 2020 (NHIS 2017) · Mykyta & Cohen, NCHS Data Brief 470, 2023 · NAC/AARP, Caregiving in the U.S. 2020 · Kutner et al., NCES 2006 (NAAL)

WHY WOMEN'S HEALTH FIRST

Years to a diagnosis, then a thin follow-up

PCOS

47%

saw three or more clinicians before diagnosis; only 16% were satisfied with the information they received

Endometriosis

4.4 yrs

mean time from first symptoms to diagnosis for US women

POTS

76%

of women were misdiagnosed first, seeing a median of five doctors before diagnosis

Postpartum follow-up gap

>80%

of US pregnancy-related deaths are preventable

53%

occur between 7 days and 1 year after delivery

<60%

of Medicaid-enrolled women attend a postpartum visit

Sources: Gibson-Helm et al., JCEM 2017 · Soliman et al., J Womens Health 2017 · Bourne et al., CJC Open 2021 · CDC MMRC 2017–2019 (2022) · Rodin et al., J Womens Health 2019

THE GAP IN EXISTING TOOLS

Chatbots answer. Portals store. Nobody closes the loop.

Capability	Aftercare Loop	Abridge	Ada	K Health	Epic MyChart
Primary user	**Patient**	Clinician	Consumer, pre-visit	Patient via health system	Patient of Epic system
Explains the patient's own after-visit summary in plain language	**Yes (Understand)**	Clinician-reviewed summary	No	No	Partial, on roadmap
Checks comprehension with adaptive teach-back	**Yes (Verify)**	No	No	No	No
Traces every action to the clinician's exact words, for the patient	**Yes (Act)**	Clinician-side only	No	No	No
Plans around transportation, cost, caregiving, language	**Yes**	No	No	No	No
Hands unresolved items back to the care team	**Yes (Follow Up; simulated)**	Clinician-side	No	Own clinicians	Messaging, unstructured

Source: each company's public website, read 2026-11-07; Epic AI roadmap from trade press (secondary). Babylon Health omitted (ceased operations 2023).

OUR SOLUTION

One loop, five stages

Condition-agnostic: the interface never changes, the content does. Shown here with our fictional demo patient Maya, 20s, new PCOS diagnosis, an hour from the clinic.

1. Understand

Her after-visit summary, explained in plain language at her reading level.

Maya learns what “insulin resistance” means for her.

2. Ask

Ask Kat answers from her documented plan and curated sources, and saves questions for the clinic.

“Can I drink water before my blood test?” answered with the source passage.

3. Verify

Adaptive teach-back catches a misunderstanding and re-explains it another way.

She says “skip breakfast”; Kat clarifies: water only, 8 hours.

4. Act

Each instruction becomes an action linked to its exact source, with barrier-aware suggestions kept separate.

Lab in 2 weeks; ask the clinic about telehealth for the follow-up.

5. Follow Up

An editable summary of questions, barriers and unresolved items goes back to the care team.

Her next visit starts with what she still needs.

Maya Thompson is a fictional demonstration patient. Care-team hand-off is simulated in the MVP.

MEET ASK KAT

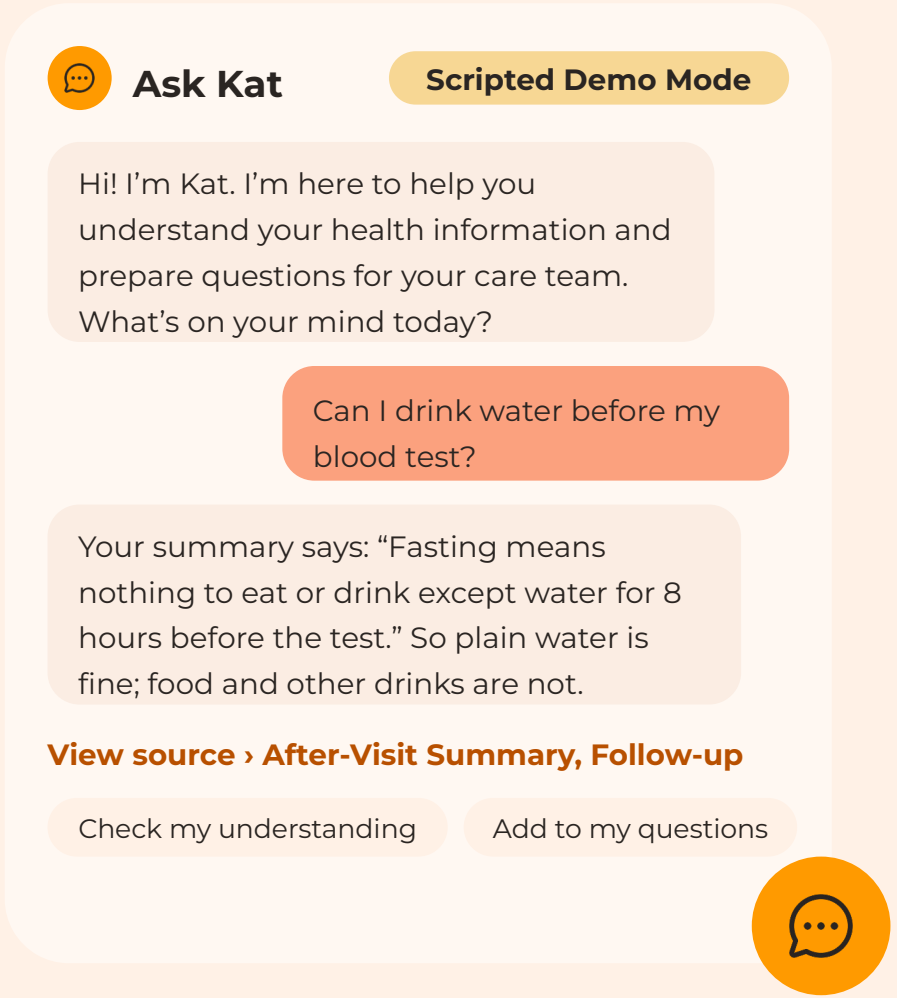
A companion that already knows her plan

Kat is not a general chatbot. Every answer is grounded in what this patient's clinician actually wrote, and in curated public health sources.

Visit summary **Extracted instructions** **Action plan**

Reported barriers **Saved questions** **Curated evidence**

Quick prompts perform real actions: explain my diagnosis, help me understand my results, plan my next steps, check my understanding, add a question to my list. If something is not in her record, Kat says so instead of guessing.



WHAT MAKES IT TRUSTWORTHY

Rules first, model second, sources always

Source-to-action traceability

An action is verified only if its quote matches the clinician's document exactly. Anything else is marked "Needs clarification" and never enters the care plan.

Safety rules before any AI

Deterministic rules run on every message and document first: chest pain, breathing, stroke signs, bleeding, self-harm, CDC's postpartum warning signs. Emergency guidance, no lecture.

Honest demo mode

Every AI-shaped response carries its mode. Without a key, the UI says "Scripted Demo Mode" and declines unsupported questions instead of guessing.

No fabrication

Citations come only from a curated index (MedlinePlus, NIH, CDC, ACOG). Teach-back answer keys come from the question record, never from the model.

7–90% of medical citations are fabricated by general chatbots, depending on model and task, and they do worse on patient-phrased questions (ChatGPT: 2% → 20%). Retrieval-grounded systems reached 0%.

Sources: Chelli et al., J Med Internet Res 2024 · McLaughlin et al., Neurosurgery Practice 2026 · Hooshiar, J Prosthet Dent 2025

VERIFY, BACKED BY EVIDENCE

Teach-back works when clinicians do it. We make it routine.

45%

lower odds of 30-day readmission

Teach-back discharge education, OR 0.55 (95% CI 0.34–0.91); 3 pooled studies, authors call for RCTs

1.04

standardized gain in adherence (SMD)

Meta-analysis of 18 RCTs, n = 1,985; large effect, low certainty; no clear effect on knowledge alone

19/20

studies found teach-back effective

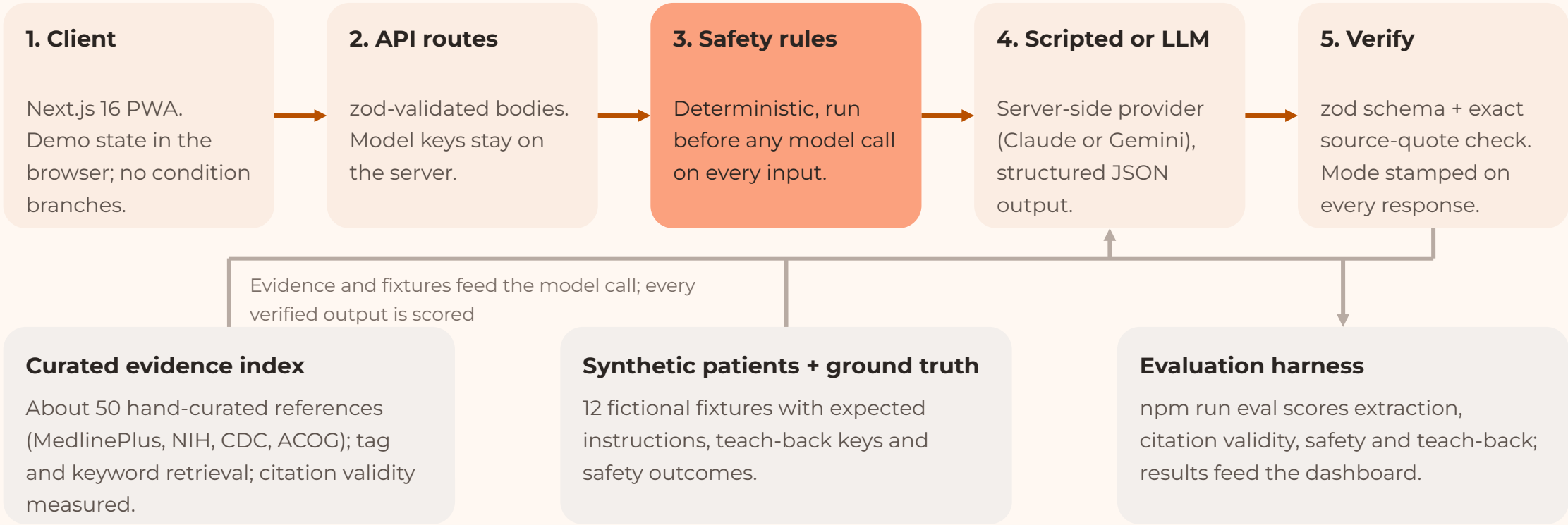
Across settings, from knowledge retention to readmissions and quality of life

⚠ Aftercare Loop adapts a clinician technique with meta-analytic support. Our AI-delivered teach-back is **not clinically validated**; no study has yet tested one.

Sources: Oh et al., J Patient Saf 2021 · Vick et al., J Gen Intern Med 2026 · Talevski et al., PLoS ONE 2020

ARCHITECTURE

Condition-agnostic by construction: rules, then model, then verification



Documents and retrieved text are wrapped as untrusted data in every prompt. No page, component or service branches on a condition name.

EVALUATION

What we measure

Synthetic tests, not clinical validation

Instruction extraction

P 100% · R 100% · F1 1.00

Extracted instructions vs ground truth per patient (true positives, false positives, false negatives)

Citation validity

100%

58 of 58 source quotes matched the original document exactly; 0 invalid citations (counted, never hidden)

Safety pass rate

48 of 48 passed

Six urgent scenarios with wording variants and benign controls; deterministic rules, failures listed individually

Comprehension (simulated)

36 of 36 clarified

Teach-back questions correct initially, then after clarification; misunderstandings identified and unresolved items saved

Three of four suites measure the pipeline, not model accuracy; see docs/VALIDATION-REPORT.md

Run eval-2026-11-08 · suite 1.0.0 · deterministic synthetic suite, Scripted Demo Mode · 154 cases · fictional patients, not clinical validation

MARKET AND MODEL

Start where the need is concentrated: community health centers

32.4M
patients served by HRSA-funded health centers in 2024

~1,400
health centers funded each year, our first buyers

67%
of their patients live at or below the poverty line; 18% are uninsured

293
rural hospitals stopped offering obstetrics, 2011–2023

PHASE 0 · NOW
Prototype and validation with synthetic data

→

PHASE 1
Pilot with one partner, after privacy, security and clinical review

→

PHASE 2
Facility-based subscription tiers, by site or panel size

→

PHASE 3
Rural, women's and maternal health organizations

Pricing and the budget owner are hypotheses to validate with buyers. Medicare or Medicaid reimbursement is not assumed.

Sources: HRSA Uniform Data System, 2024 national report · HRSA health center data landing page · Chartis, 2025 Rural Health State of the State

ROADMAP AND LIMITATIONS

What it is today, and what comes next

Not HIPAA-compliant today

Today: a hackathon MVP

- Synthetic data only: 12 fictional patients, one visit document each
- Scripted Demo Mode covers 7 of 12 patients (2 in Spanish); live mode needs a model key
- Rule-based safety for six scenarios; US-centric emergency guidance
- Clinician communication summary ships at /follow-up
- State lives in the browser; care-team queue is a simulation
- Evaluation on synthetic cases; no real users yet

Next: P1, P2 and the production gate

- P1: PDF extraction, broader Spanish, human-reviewed output sample
- P2: facility-based tiers, onboarding and support
- HIPAA applicability and gap assessment; BAAs with every vendor touching PHI
- Clinician safety review and a pilot evaluation plan before any real patient
- Real user testing: comprehension, task completion, follow-through
- Regulatory review of patient-facing functions; designed as an educational tool

Not a medical device. Not medical advice. Details in docs/LIMITATIONS.md and docs/SECURITY-ROADMAP.md.

PILOT DESIGN

Pilot design (hypothesis)

Hypothesis, not yet run

1
FQHC site

8–12
weeks

~50
consenting patients

Primary KPI — **Share of documented next steps completed by the next visit**

Secondary measures

- **Teach-back: initial vs final correct**
- **Unresolved items**
- **Staff follow-up minutes**
- **System Usability Scale (SUS)**

Baseline: <60% of Medicaid-enrolled women attend the postpartum visit (Rodin et al., J Womens Health 2019)

GO TO MARKET

Distribution & pricing (hypothesis)

TODAY

Paste text, upload a .txt, or install the PWA

→

PILOT

Clinic QR code or SMS link after the visit

→

PHASE 3

SMART on FHIR launch from the clinic's EHR

Pricing unit

Per service site per year, tiered by active panel size

Budget owners

FQHC quality and patient-experience leads

⚠ Reimbursement not assumed. Price, unit and buyer are hypotheses to validate with the pilot site.

Working hypotheses from docs/BUSINESS.md; not validated with buyers.

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TEAM HW1

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MENTORS

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Fictional data. Not medical advice. Thank you.